

Figure 1: WireShark running

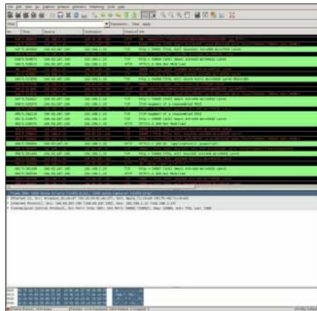


Figure 2: WireShark capturing data

Figure 3). There you can select the Interface (*en0*), see your IP address (192.168.1.10), apply any *Capture Filter* (in the image, there is none), put your network card in *promiscuous mode*, and save captured data to one or more files. When capturing lots of data, it is considered a good practice to first save and then examine captured network traffic. When you put your network card in *promiscuous mode*, you allow the network device to catch and read every network packet that arrives, even if the receiver is another device on the network. You can also choose to stop packet capturing after a given number of network packets, a given period of time, or a given amount of data (in bytes).

WireShark allows you to read and analyse captured network data from a large number of file formats including *tcpdump*, *libpcap*, Sun's *snoop*, HP's *nettl*, K12 text file, etc. This means that you can read almost every kind of captured network data

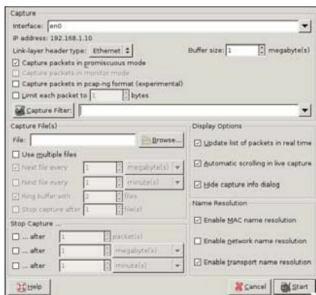


Figure 3: WireShark Capture options



Figure 4: Supported file formats

with WireShark, while new file formats are frequently added. It is more likely that WireShark cannot read a file due to invalid packet types than an incompatible format! Similarly, WireShark allows saving captured network data in a variety of formats, visible in Figure 4. You can even use WireShark to convert a file from one format to another.

## WireShark display filters

The amount of network data that WireShark may display can be too much for a human to watch and understand, especially on very busy

networks. Usually, when using WireShark, we want to examine a given problem or situation, or even watch for unusual network activity. WireShark allows us to filter network data for specific types of traffic during capture, avoiding creating huge capture files—but there are also display filters, which tell WireShark to display packets that really matter, while capturing everything, so that you look at fewer packets and easily find what you want.

Generally speaking, display filters are considered more practical and versatile than capture filters, because most of the time, you don't know in advance what you want to examine. Nevertheless, using capture filters can save time and disk space; that is the main reason for using them.

WireShark tells you when a Display Filter is syntactically correct—when the background turns light green, then the filter is syntactically correct; and when the syntax is erroneous, the background is pink. You can see both cases in Figure 5. The result of a logically inaccurate (yet syntactically correct) filter at capture time is no captured data—so you may recognise this error the hard way.